RATING METHODOLOGY

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Rating Methodology

Homebuilding and Property Development

This rating methodology replaces the *Homebuilding and Property Development* methodology published on 21 October 2022. We have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Business Profile factor, we have renamed the Market Conditions subfactor to Market Characteristics. In Profitability and Efficiency, we have replaced the Gross Margin subfactor with EBIT Margin. We have changed some factor weights and scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. We have also changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in the construction and sale of single-family and multifamily housing, including large-scale residential apartments. This methodology also applies to companies primarily engaged in the construction and sale of commercial properties.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

We rate companies that are primarily engaged in the ownership and operation of commercial properties for long-term investment using our methodology for REITs and other commercial real estate firms.[1] We rate companies that generate their revenue or operating cash flow from the construction or refurbishing of buildings for commercial purposes, such as offices or warehouses, or for public purposes, such as schools, hospitals or government buildings, using our construction methodology. We rate companies primarily engaged in the production, sale and distribution of building materials using our building materials methodology.

Sector overview

Some homebuilding and property development companies engage in real estate investments to generate rental income, which is a highly capital-intensive business. This leads to a generally higher leverage for companies that have meaningful real estate investments. However, the high margins and recurring nature of rental income can mitigate the impact of increased leverage from such investments. We assess the exposure to real estate investment qualitatively, such as the benefits arising from business diversification, stable and recurring nature of rental income, and higher margins to support property development. We also assess the capital needs to support these real estate investments.

Some homebuilding and property development companies have captive finance subsidiaries that provide mortgage financing for customers. The debt of these subsidiaries is typically non-recourse to the homebuilding and property development parent and secured by customer mortgages. The debt generally amortizes quickly, as the mortgages are typically sold within a 60-day period, with proceeds used to pay down the captive finance subsidiary's debt.

Captive finance subsidiaries can constrain a homebuilding and property development company's financial flexibility. This stress can result from the finance operation's need for supplementary equity capital and liquidity, which is more likely to occur in periods of market stress, when a company's core operations are also affected. We typically consider the likelihood and cost of the support that the company may extend to its captive finance subsidiary. For details on how we assess the credit impact of a captive finance subsidiary on a homebuilding and property development company, see our methodology for rating captive finance subsidiaries of nonfinancial corporations.[2]

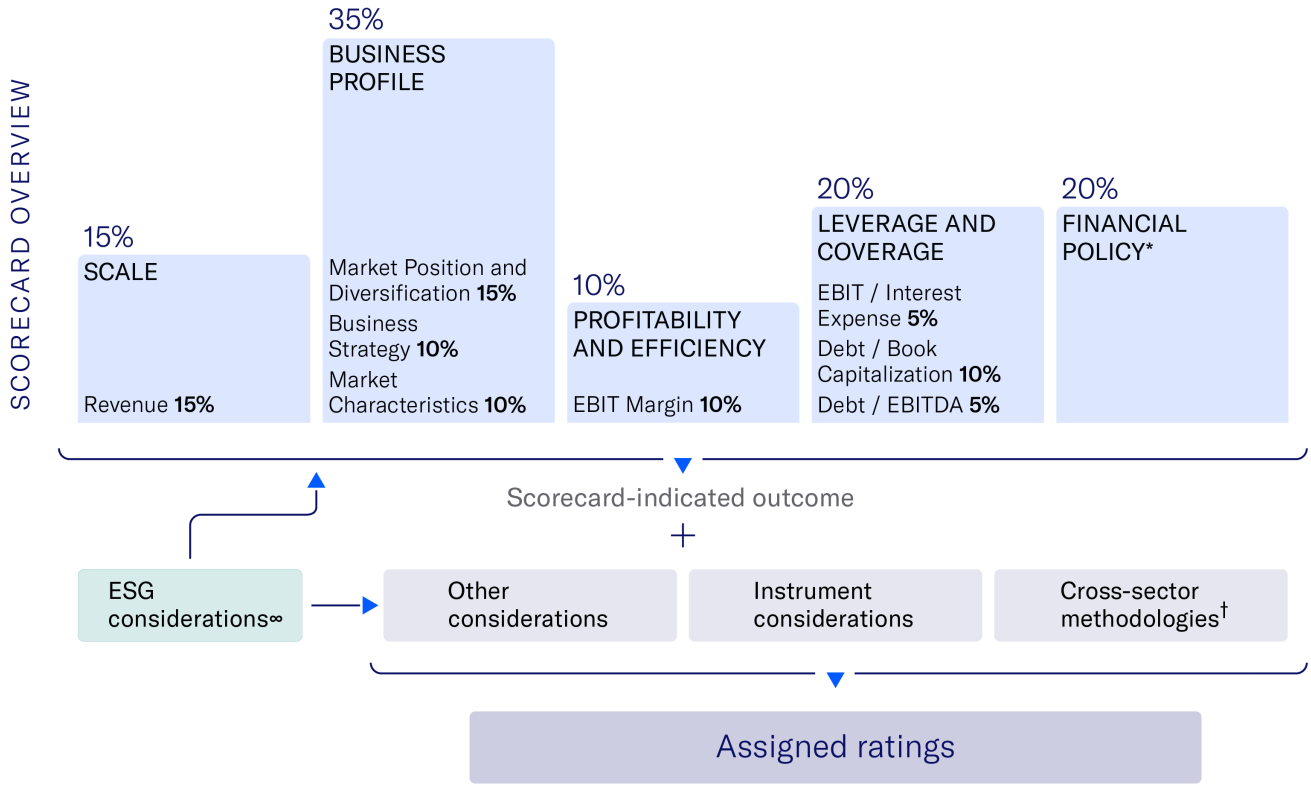
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the homebuilding and property development rating approach



* This factor has no subfactors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Homebuilding and property development scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (15%)									
Revenue (USD Billion) [1]	15%	≥ $100	$50 - $100	$25 - $50	$10 - $25	$4 - $10	$1.5 - $4	$0.5 - $1.5	< $0.5
Factor: Business Profile (35%)									
Market Position and Diversification	15%	Very strong market position in multiple countries; and products are offered in a very wide variety of property types and customer segments.	Very strong market position in many large markets in a country; and products are offered in a wide variety of property types and customer segments.	Strong market position in many large markets in a country; products are offered in a wide variety of property types and customer segments.	Strong market position in many markets in a country; products are offered in many property types and customer segments.	Solid market position in several markets in a country; products are offered in several property types and customer segments.	Modest market position in a few markets in a country; products are offered in a few property types or customer segments.	Weak market position in a single market; company relies on a very small number of product types or customer segments.	Very weak market position in a small market; or company relies on a single product type or a single customer segment.
Business Strategy	10%	Extremely conservative inventory management and funding strategy, with a very strong land bank and housing inventory that has no exposure to price declines because of mitigation measures.	Very conservative inventory management and funding strategy, with a strong land bank and housing inventory that has minimal exposure to price declines because of mitigation measures.	Conservative inventory management and funding strategy, with a strong land bank and housing inventory that has low exposure to price declines because of mitigation measures.	Somewhat conservative inventory management and funding strategy, with a solid land bank and housing inventory that has modest exposure to price declines because of mitigation measures.	Inventory management and funding strategy is balanced, with issuer having somewhat elevated exposure to price declines from a land bank and housing inventory that are sized to meet operational requirements.	Inventory management and funding strategy is somewhat speculative, with issuer having sizable exposure to price declines from a land bank and housing inventory that are moderately larger than operational requirements.	Inventory management and funding strategy is speculative, with issuer having material exposure to price declines from a land bank and housing inventory that are considerably larger than operational requirements.	Inventory management and funding strategy is very speculative and largely financed with debt, with issuer being highly vulnerable to price declines from a land bank and housing inventory that are excessive relative to operational requirements.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Market Characteristics	10%	Extremely strong and sustainable end-user demand that is expected to exceed supply; extremely long track record of solid and stable property price growth, which is expected to continue; operates in a regulatory environment with a long and consistent track record of being extremely predictable, stable and favorable.	Very strong and sustainable end-user demand that is expected to exceed supply; very long track record of solid and stable property price growth, which is expected to continue; operates in a regulatory environment with a consistent track record of being extremely predictable, stable and favorable.	Strong and sustainable end-user demand that is expected to exceed supply; long track record of solid and stable property price growth, which is expected to continue; operates in a regulatory environment with a track record of being very predictable, stable and favorable.	Strong end-user demand that is expected to exceed supply; modest track record of solid and stable property price growth, which is expected to continue; operates in a regulatory environment with a track record of being somewhat stable and favorable.	Stable market dynamics, with end-user demand and supply expected to be largely in balance; property prices are expected to remain largely stable, with limited effect from speculative activity in the market; operates in a regulatory environment with a track record of being somewhat stable but neutral.	Stable market dynamics, but supply is expected to exceed end-user demand; property prices are expected to remain somewhat volatile, with some effect from speculative activity in the market; operates in a regulatory environment with a track record of being neutral, but with some potential for adverse change.	Weakening market dynamics, with supply expected to exceed end-user demand; property prices are expected to be volatile, with a material effect from speculative activity in the market; operates in a somewhat unpredictable regulatory environment with adverse change likely.	Weak market dynamics, with supply exceeding end-user demand on a sustained basis; property prices are expected to decline steeply, with high levels of speculative activity in the market, operates in a highly unpredictable regulatory environment with significant adverse change occurring or highly likely.
Factor: Profitability and Efficiency (10%)									
EBIT Margin [2]	10%	≥ 60%	35% - 60%	25% - 35%	20% - 25%	15% - 20%	10% - 15%	5% - 10%	< 5%
Factor: Leverage and Coverage (20%)									
EBIT / Interest Expense [3]	5%	≥ 20x	15x - 20x	10x - 15x	5x - 10x	3x - 5x	1.25x - 3x	0.5x - 1.25x	<0.5x
Debt / Book Capitalization [4]	10%	≤ 20%	20% - 25%	25% - 30%	30% - 40%	40% - 50%	50% - 65%	65% - 80%	> 80%
Debt / EBITDA [5]	5%	≤0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4.5x	4.5x - 6.5x	6.5x - 8.5x	> 8.5x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $150 billion. A value of $150 billion or better equates to a numeric score of 0.5. The Ca endpoint value is $0. A value of $0 equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 85%. A value of 85% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 30x. A value of 30x or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 0%. A value of 0% equates to a numeric score of 0.5. The Ca endpoint value is 100%. A value of 100% or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative EBITDA value.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[3] We may also make nonstandard adjustments that are specific to a particular company. See the appendix for examples of nonstandard adjustments that apply to this sector.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (15% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (35% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Market Position and Diversification

The sustainability and strength of a homebuilding and property development company's market position provide important indications of its competitive strength, brand recognition, ability to attract talent and capital, and ability to withstand market volatility.

Homebuilders and developers with a strong market position are generally less susceptible to downturns in economic growth and consumer spending. They may find it easier to attract and retain skilled employees without paying excessive compensation and may not need to devote as many resources to promotional sales practices as developers that have a weaker market position. Companies with a strong market position may also have greater access to capital at a competitive cost. Conversely, companies with a weak market position could have their access to funding interrupted when credit conditions tighten, potentially increasing refinancing and liquidity risks.

A diverse mix of geographic markets, property types and customer segments is also important because it reduces a developer's vulnerability to economic shocks, regulatory issues, cyclicality or other adverse market developments that could cause earnings to

erode. Operating in large and diverse geographic markets and offering a wide variety of property types to many customer segments can lessen the impact of local market cyclicality and supply and demand shifts, regulatory changes, competitors' actions and other developments that could significantly curtail operations or profitability for companies in this sector. Companies with these characteristics also have increased flexibility to adjust their geographic focus or product offerings in response to changes in market demand, demographics, and economic and regulatory environments. Conversely, concentration in a limited number of geographic areas, products types and customer segments can expose a company to losses related to changes in those markets.

Business Strategy

Our assessment of a company's business strategy provides important indications of management's operational risk appetite and the company's exposure to volatile land and home prices. The homebuilding and property development sector is exposed to a high level of cyclicality, which can have a material impact on companies' profitability and financial strength due to the large amounts of capital needed to purchase, develop and hold large amounts of land and properties under development. In addition, market conditions may deteriorate significantly during the holding period, i.e., after these investments are made and before homes or commercial properties are sold to customers. Most companies in this sector have high working capital needs that can lead to large swings in cash flow. Companies with prudent risk management and funding strategies are less susceptible to earnings and cash flow volatility arising from changes in business, funding and regulatory environments.

Market Characteristics

The characteristics of the markets and the regulatory environment in which a homebuilder or property developer operates are important because they indicate the strength, stability and sustainability of demand and property prices. They also provide important indications of the potential impact of regulatory changes on market supply and demand, on operational constraints, and on the cost and availability of funding.

Strong and sustained underlying demand, coupled with stable and predictable supply, provide a clearer view of a company's home prices, profitability and cash generation. Volatile supply-and-demand dynamics increase uncertainty in revenue and cash flow forecasts, increase risks related to the execution of management's business plans and may result in significant land or inventory impairments and reduced profitability.

Regulation of the homebuilding and property development sector can affect companies in many ways, including through changes to demand or supply of land or property, increases to the cost of operating, and restrictions in the availability or cost of financing. Predictable and favorable regulatory change can mitigate the impact of other demand and supply drivers, such as macroeconomic conditions or speculative activity. Conversely, unpredictable or adverse regulatory change increases business execution risks in this sector and can magnify the adverse impacts of changes in market characteristics.

How we assess it for the scorecard

We score this factor based on three qualitative subfactors: Market Position and Diversification, Business Strategy, and Market Characteristics.

We generally do not expect a given company to match each attribute listed for a given scoring category for each subfactor. We typically assign a subfactor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on a subfactor score.

Market Position and Diversification

We assess market position primarily based on a company's market share and position in each of its markets. We consider publicly available company and market data, and we make a qualitative assessment of characteristics such as regional footprint and diversification, product type and customer segment, reputation, and track record of execution.

Our assessment of diversification is holistic and considers the extent that geographic, product or customer diversity can each reduce a company's exposure to adverse events. The extent to which certain adverse events have a widespread impact limits the benefits of diversification.

We assess geographic diversification primarily based on the extent to which a company's revenue is spread across cities, metropolitan areas, regions, provinces or countries. We assess product type and customer segment diversification based on the spread of revenue across different product types (e.g., detached single-family homes, town homes, apartments, planned communities and commercial property) and the variety of customer segments (e.g., first-time buyers, move-up buyers/upgraders, luxury buyers and investors). We assess product type and customer segment diversification of developers of primarily commercial properties based on the spread of revenue across different product types within the commercial property market.

A homebuilding and property development company with significant market share, that is recognized as a market leader in multiple markets, and that offers a variety of property types to several customer segments typically receives higher scores for this subfactor than a company whose revenue is concentrated in a single market and that has one product type only.

Business Strategy

We assess management's strategy for managing risks related to the significant inventory investment required to buy land and build properties. We also consider the risk that demand and sales prices for the finished product will be lower than anticipated.

Companies that use debt to acquire land that is not earmarked for upcoming operational purposes are likely to score lower for this subfactor, as are companies that build large numbers of properties without home purchase orders or in markets where demand for those properties is unproven. Conversely, companies that have mitigated inventory risks through the use of land purchase options (rather than outright buying of land) and through the building of properties based on customer orders are likely to score higher for this subfactor. Companies with large legacy land banks acquired at a low cost may also score higher for this subfactor because they have operational flexibility in land bank replenishment.

Market Characteristics

We assess market characteristics based on the level of end-user demand relative to supply, the stability of property prices and the impact of speculative activity, and the regulatory environment.

In assessing demand and supply, we consider macroeconomic indicators, such as economic growth and employment data; industry data, such as housing starts, housing completions, mortgage activity and affordability; and demographic trends, such as population change and urbanization, to form a two-to-three-year outlook.

Our outlook for property prices is informed by historical trends and our assessment of demand and supply dynamics, as well as potential external drivers such as interest rates and regulatory changes. Our forward-looking view of the regulatory environment is largely based on the historical track record and on regulatory pronouncements. The regulatory environment can encompass a wide spectrum of areas, including land zoning and building permitting, construction controls and standards, taxes and stamp duties, mortgage lending standards, restrictions on non-local and second-home buyers, caps on selling prices, and bank prudential requirements that affect the availability of credit.

A company operating in markets with strong and consistent end-user demand that is expected to exceed supply, where prices are expected to grow solidly, and where the regulatory environment is stable and favorable, is likely to receive a higher score for this subfactor. Companies that are exposed to markets with weak demand and declining prices and that operate in an unpredictable regulatory environment are likely receive a lower score.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[4]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (20% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: EBIT/Interest Expense, Debt/Book Capitalization and Debt/EBITDA.[5]

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

Debt / Book Capitalization

We use total debt to book capitalization.

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand business lines, achieve economies of scale, consolidate market positions or seek access to new technology.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type

of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[6]

Homebuilding and property development companies are exposed to environmental risk in the form of physical climate risks, in which events related to climate change can have a negative and lasting impact on a company's assets. Companies that develop assets with construction cycles exceeding 12 months or own long-dated land banks in locations subject to physical climate risks such as flooding and typhoons can be exposed to higher construction costs or asset impairment risk. The ongoing requirement to exploit land resources for business expansion also exposes companies to natural capital risks, where the natural resources of a company can be damaged, and associated environmental regulation compliance costs related to land preservation.

Changing demographic and societal trends can introduce social risks that may have an impact on demand for property and on the earnings of homebuilding and property development companies. In certain markets such as mainland China, change in regulatory requirements can also disrupt the business operations of companies. Other social considerations include customer relationships, information protection and responsible production. Homebuilding and property development companies' commitments to deliver quality products to customers on time and the need to safeguard sensitive customer data may expose them to customer relations risks. The construction process and supply chain may lead to risks associated with achieving responsible production. For example, reduced transparency into the land procurement process in some emerging markets may increase the risk of bribery and corruption.

Governance considerations include the ownership and control of homebuilding and property development companies. Concentrated ownership and voting control may exert a potentially negative influence on corporate performance and credit outcomes because owners may seek to extract private benefits at the expense of other stakeholders. Companies in this sector may also be exposed to key-person risk, in which dependence on a single individual or a limited group of executives can adversely impact operations, especially

in the absence of a succession plan. Related-party transactions may indicate a governance weakness and create conflicts of interest, reputational damage and, in severe cases, can impair the ability of the developer to obtain external financing. Complex organizational structures may expose a developer to governance risks because significant cross-shareholdings or frequent changes in organizational structure may increase the risk of a misallocation of funds, as well as reduce corporate transparency.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[7]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Joint ventures

Joint ventures (JVs) can be a credit strength by providing earnings diversification and other means of capital access for homebuilders and property developers. But they can also be complex structures and may create varying levels of opacity and governance risks. In addition, the earnings quality of a homebuilding and property developer can be diminished if these structures generate a large proportion of earnings.

As a result, JVs provide a mix of credit-positive and credit-negative characteristics, based on the transaction specifics and the overall contribution of these transactions to a developer's revenue stream. The use of off-balance sheet structures such as JVs can reduce financial transparency and can indicate a company's appetite for growth despite capital constraints. The use of joint ventures also generally restricts management's control over projects, as the agreement of partners is typically required for major decisions.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity

stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

For example, the scorecard does not include tangible net worth, but it can often provide additional analytical insight in our assessment of the scale of homebuilders and property developers. Unlike other industries, in which highly leveraged companies with large amounts of negative tangible net worth can operate with a modicum of success, the homebuilding and property development industry typically needs large amounts of capital to purchase, develop and hold large amounts of land and work-in-process inventories. As a result, a company with negative or very low tangible net worth may have an untenable capital structure.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic

incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[8]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3
Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4
Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[10]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[11]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[12]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[13] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Appendix: Nonstandard adjustments to financial statements

In addition to making standard adjustments to a company's financial statements, we may also make nonstandard adjustments to better reflect underlying economics, improve comparability with peer companies and reflect estimates or assumptions we consider to be more appropriate for credit analysis.

We make nonstandard financial statement adjustments based on public information, although we may consider private information in our ratings. We may apply a materiality threshold to determine whether or not to make a nonstandard adjustment to a particular company's financial statements.

For companies in this sector, we make the following nonstandard adjustments to certain financial metrics:

We adjust gross profit, EBIT and EBITDA to exclude interest charged to cost of goods sold (CGS). Because we consider capitalized interest as a cost of financing (i.e., an interest expense) that should be expensed when it is incurred, we adjust interest expense by reclassifying capitalized interest (including interest capitalized into inventory and interest capitalized into property plant and equipment) as interest expense. We also remove interest charged to CGS from CGS to avoid double-counting. If a company's financial statements do not consistently disclose interest charged to CGS, we use total capitalized interest as a proxy for interest charged to CGS. For clarity, the adjustment to include interest capitalized into property, plant and equipment in interest expense is a standard adjustment.

We further adjust gross profit, EBIT and EBITDA to exclude land appreciation tax, which we consider as a taxation expense, in addition to excluding land impairment charges, which is a standard adjustment.

We adjust EBIT and EBITDA to exclude unremitted equity income or losses from off-balance sheet JVs and include in our adjustment dividends received from off-balance sheet JVs.

Our homebuilding and property development financial ratios typically exclude the revenue, costs and debt of captive financial subsidiaries. Without these adjustments, peer comparisons of data and ratios are difficult because not all companies have finance operations.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

12 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

13 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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